

Hazidien Ramadhan Utomo

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Summary

Urban and Regional Planning graduate from Institut Teknologi Sepuluh Nopember (ITS) who uses GIS and spatial data to support planning decisions. Built a LiDAR-based rooftop solar potential model covering 31 districts in Surabaya, validated against SolarGIS (7.36% relative error), and worked on land-use change simulation to 2043 with GeoSOS-FLUS and deforestation analysis in Google Earth Engine. Experienced in ArcGIS, QGIS, LiDAR processing, and remote sensing through a renewable energy spatial analysis internship and a role as assistant lecturer in planning analysis methods. Also brings cadastral mapping, field survey, and stakeholder engagement experience from an agropolitan planning studio and the MRT Surabaya project, along with team coordination from student organization leadership.

Education

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

B.Sc. in Urban and Regional Planning,

Aug 2022 – Aug 2026

GPA: 3.36/4.00

Achievement:

- 3rd Place Winner of the Infographic Research Competition – GIS THEMATICAL COMPETITION National Level
- 1st place in Football regional competition of Surabaya city Trofeo UINSA

Work Experiences

Assistant Lecturer – Planning Analysis Methods / Metode Analisis Perencanaan

Aug 2025 - Dec 2025

Laboratory of Transportation and Spatial Analysis, Department of Urban and Regional Planning, ITS

Surabaya, Indonesia

- Delivered laboratory sessions and guided students through **quantitative, qualitative, and GIS-based planning analysis**, incorporating **AI-assisted learning tools** to support clear and structured explanations.
- Presented and explained technical concepts, including **spatial regression, clustering, and spatial autocorrelation**, in an accessible and engaging manner.
- Facilitated discussions, responded to students' questions, and provided feedback to support **effective communication, understanding, and problem-solving**.

PT Greenova Daya Prima

Jul 2025 - Aug 2025

Spatial Analysis Assistant – Renewable Energy (Internship)

Surabaya, Indonesia

- Assessed rooftop solar potential by integrating **building characteristics, land use, LiDAR-derived data, and solar radiation to support urban energy planning**.
- Supported the Surabaya Building Energy Baseline Study by integrating building, electricity consumption, land-use, and energy indicator datasets.
- Assisted coordination and validation of survey data covering **233 residential and non-residential buildings**.

Research Assistant – Land-Use Change Modelling

Jan 2025 - Feb 2025

Laboratory of Transportation and Spatial Analysis, Department of Urban and Regional Planning, ITS

Surabaya, Indonesia

- Conducted **land-use change modelling using GeoSOS-FLUS** based on 2018 and 2023 land-use data and developed scenarios for 2023–2043 using ANN and Markov Chain analysis.
- Evaluated model performance using **Kappa statistics and Figure of Merit (FoM)** to assess spatial agreement and the results of land-use change simulations.

British Embassy Jakarta

Jun 2024 - Nov 2024

Surveyor MRT Surabaya Project

Surabaya, Indonesia

- Conducted **household, employee, and administrative surveys**, communicating directly with respondents to explain survey objectives, gather information, and ensure accurate field data collection.
- Coordinated with **survey teams and fellow surveyors** during field activities, supporting effective communication, task coordination, and consistent data collection procedures.
- Performed **traffic counting and vehicle speed surveys** and supported the analysis of mobility patterns and transportation planning scenarios.

Project Experiences

Thesis Research

Sep 2025 – Jul 2026

- Designed a **GIS-based spatial analysis methodology** to assess **rooftop solar energy potential across 31 districts in Surabaya** by integrating building footprints, land use, LiDAR-derived rooftop data, and solar radiation
- Conducted **LiDAR-based Area Solar Radiation modelling in ArcGIS** and validated the results against SolarGIS, achieving an **RMSE of 143.40 kWh/m²/year (7.36% relative error)**.

Spatio-Temporal Deforestation Modeling Project – West Sumatra

Nov 2025 – Dec 2025

- Developed **spatio-temporal deforestation analysis workflows using Google Earth Engine (GEE)** and multi-temporal Landsat imagery from **2009–2024** for environmental assessment.

- Evaluated and refined **AI-assisted GEE workflows** by reviewing analytical logic, parameter choices, and spatial results to improve the reliability of deforestation analysis.

Regional Planning Studio — Agropolitan Area Development

Aug 2024 – Dec 2024

- Conducted regional assessment of **agricultural potential, infrastructure needs, leading commodities, and service centres**, supported by **cadastral mapping and spatial analysis**.
- Engaged in **stakeholder discussions and Focus Group Discussions (FGDs)** to identify local conditions, development needs, and planning priorities.

Leadership & Organizational Experince

UKM Sepakbola ITS

Jan 2024 - Dec 2024

Head of the Football Division

- Coordinated **training activities, member development, and team preparation** for regional football competitions
- Managed team coordination and served as **acting coach** during selected training sessions and competitions.

Aksi Sepuluh Nopember 2023

Aug 2023 - Dec 2023

Vice Chairman I – Campus Orientation Event

- Coordinated and supervised **events, equipment, logistics, and operations** during the preparation and execution of a large-scale campus orientation event.
- Assisted in **planning, team coordination, and decision-making** throughout the event and contributed to its successful implementation.

Certification

- **Anthropic:** Claude 101 – AI Collaboration & Responsible AI Use (2026)
- **Aksara Lab Indonesia:** Intensive Bootcamp – LiDAR Data Classification & Contour Generation (2025)
- **Geoaccess Indonesia:** Thematic GIS Training – Land Cover Change Prediction using QGIS & GeoSOS-FLUS (2025)
- **Geoaccess Indonesia:** Thematic GIS Training – Flood Modeling Based on Land Use Change & Prediction (2024)
- **Geoaccess Indonesia:** Basic & Thematic Google Earth Engine (GEE) Training – Deforestation Impact on Air Pollution Analysis (2024)

Core Skills

- Analysis & Planning:**
 - Spatial reasoning, causal analysis, scenario-based assessment, uncertainty awareness, model evaluation, cadastral mapping
- GIS & Spatial Modeling:**
 - ArcGIS · QGIS · Google Earth Engine · AutoCAD · Remote Sensing · LiDAR Processing · Spatial Modelling · GeoSOS-FLUS
- Project & Stakeholder:**
 - Project Coordination · Stakeholder Engagement · Data Collection & Validation · Technical Communication
- AI & Geospatial Automation:**
 - AI-assisted WebGIS, QGIS–MCP integration, LLM-based tool integration, AI-assisted coding, AI–GIS Integration
- Professional:**
 - Analytical Problem-Solving · Critical Thinking · Team Collaboration · Leadership
- Languages:**
 - Indonesian (Native) · English (Professional Working Proficiency) · Javanese (Native)